Linear Regression Multiple Variables

Predict House Price given House Size

House Price

House Size (double) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ House Price (double)

Predict House Price given House Size

House Price

House Size (double) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ House Price (double)

House Price

House Size (double)
N. Bedrooms (double)
N. Floors (double) →

```
B ml_model(A input){
    ....
    return output;
}
```

→ House Price (double)

Multiple Features

Size in feet^2	Number of bedrooms	Number of floors	Age of home in years	Price ($) in $1000's
2104	5	1	45	460
1416	3	2	40	232
1534	3	2	30	315
852	2	1	36	178
...	...	...	...	...

x_j = j^{th} feature

n = number of features

$\vec{x}^{(i)}$ = features of ith training example

$x_j^{(i)}$ = value of feature j in i^{th} training example

Multi Linear Regression

Model:

$$f_{w,b}(\mathrm{x}) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

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$$f_{w,b}(\mathrm{x}) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

$$f_{\vec{\mathrm{w}},b}(\vec{\mathrm{x}}) = \vec{\mathrm{w}} \cdot \vec{\mathrm{x}} + b =$$

Gradient Descent for Multi Linear Regression

Parameters $w_1, \cdots, w_n$

b

Model $f_{\vec{w},b}(\vec{x}) = w_1 x_1 + \cdots + w_n x_n + b$

Cost function $J(w_1, \cdots, w_n, b)$

Gradient descent

repeat {

$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(w_1, \cdots, w_n, b)$

$b = b - \alpha \frac{\partial}{\partial b} J(w_1, \cdots, w_n, b)$

}

Gradient Descent for Multi Linear Regression

One feature

repeat {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)$$

simultaneously update w, b

}

n features ($n \geq 2$)

repeat {

$$w_1 = w_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right) x_1^{(i)}$$

}

Gradient Descent for Multi Linear Regression

One feature

repeat {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)}\right)$$

simultaneously update w, b

}

n features ($n \geq 2$)

repeat {

$$w_1 = w_1 - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right) x_1^{(i)}$$

$$w_n = w_n - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right) x_n^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{\vec{w},b}\left(\vec{x}^{(i)}\right) - y^{(i)}\right)$$

simultaneously update
w_j (for $j = 1, \cdots, n$) and b

}

Notebook: Multiple Linear Regression

Session3_Multiple_Linear_Regression.ipynb

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1 Introduction

1.1 Goals

- Extend our regression model routines to support multiple features
  - Extend data structures to support multiple features
  - Rewrite prediction, cost and gradient routines to support multiple features
  - Utilize NumPy `np.dot` to vectorize their implementations for speed and simplicity

1.2 Tools

In this exercise, we will make use of:

- NumPy, a popular library for scientific computing
- Matplotlib, a popular library for plotting data

Exercise 1: Feature Scaling

6 Feature Scaling

6.1 Runing Gradient Descent without Feature Scaling

```
# initialize parameters
initial_w = np.zeros_like(w_init)
initial_b = 0.
# some gradient descent settings
iterations = 10000
alpha = 0.0000001
# run gradient descent
w_final, b_final, J_hist = gradient_descent(X_train, y_train, initial_w, initial_b,
                                                    compute_cost, compute_gradient,
                                                    alpha, iterations)
print(f"b,w found by gradient descent: {b_final:0.2f},{w_final} ")
m,_ = X_train.shape
for i in range(3):
    print(f"prediction: {np.dot(X_train[i], w_final) + b_final:0.2f}, target value: {y_train[i]}")
```

Exercise 1: Explore the effect of feature scaling in Gradient Descent

Perform the following experiments:

- Observe how the learning rate has different effects before and after normalization.
- Observe how quickly gradient descent converges depending of the learning rate before and after normalization.

Feature Engineering

$$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x_2 + b$$

Feature Engineering

$$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x_2 + b$$

$$area = frontage \times depth$$

$$x_3 = x_1 x_2$$

$$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x_2 + w_3 x_3 + b$$

Feature engineering:
Using intuition to design new features, by transforming or combining original features.

Exercise 2: Feature Engineering

7 Feature Engineering

Let us introduce in our training data sample a new feature "size_per_bedrrom":

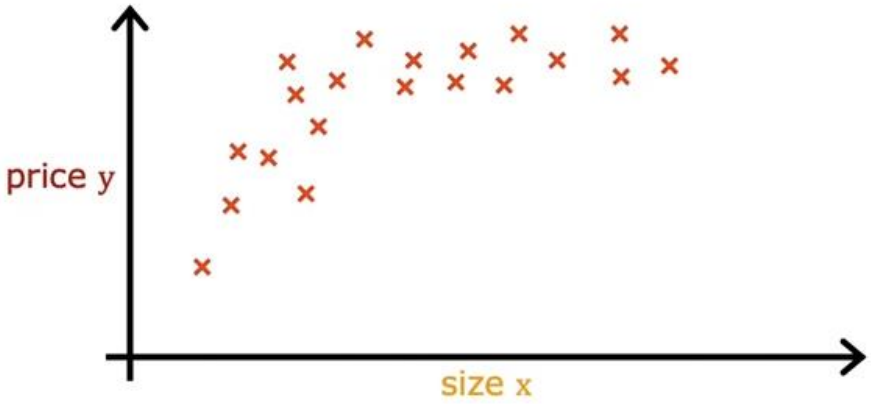
$$\text{size_per_bedrrom} = \frac{\text{size}}{\text{number of bedrooms}}$$

```
# Create a larger matrix
X_train_extended = np.zeros((X_train.shape[0],X_train.shape[1]+1))

# Copy the training data to the new matrix
X_train_extended[:,:-1] = X_train

#Add the new colum for "size_per_bedroom"
X_train_extended[:,-1] = X_train[:,0]/(X_train[:,1])

#Create new names for the variable
X_features_extended = ['size','bedrooms','floors','age', 'sizer_per_bedroom']
```

Exercise 2: Add a new feature and see what happens

- Add a new feature you think it could help to better predict the price of a house, following the same approach used to add "size_per_bedroom" feature. (i) Justify why did you decided to use this feature? (ii) Explore the training cost of the new solution an compare it with the cost of the previous solutions. Is the cost smaller?

Polyomial Regression



Exercise 3: Polynomial Features

8.1 Adding Polynomial Features

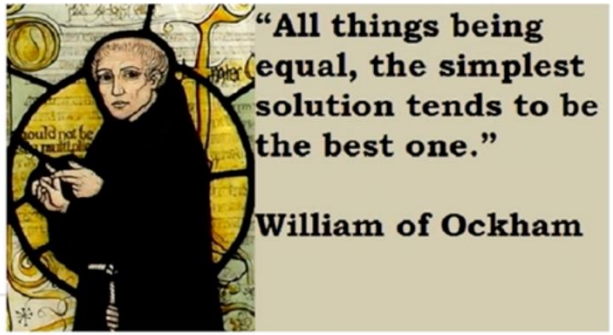
Let us add new polynomial features (i.e. $size^2$, $size^3$, etc) to our training data set and see what happens with the cost funciton after running gradient descent.

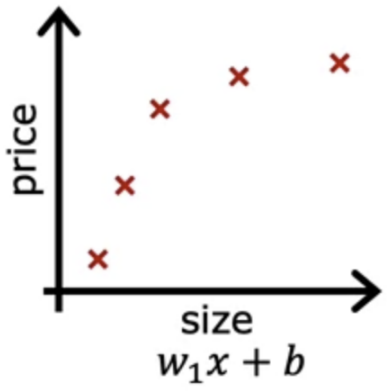
```
# Enter the degree of the new polynomial features.
degree = 2
# Create a larger matrix
X_train, y_train = load_house_data()
X_train, _, _ = zscore_normalize_features(X_train)
X_train_extended = np.zeros((X_train.shape[0],X_train.shape[1]+degree-1))
```

Exercise 3: Polynomial Features

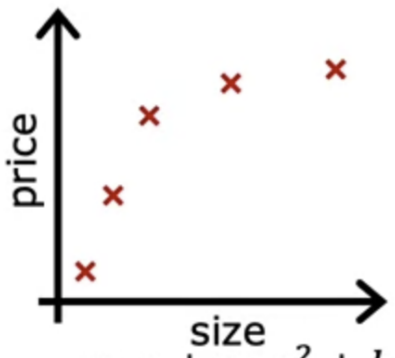
- Add polynomial features using the code above. Start with degree=2 and move to degree=20.
- Look how the Cost function is reduced.
- Do you think we are doing better by having more polynomial functions?

The problem of Overfitting

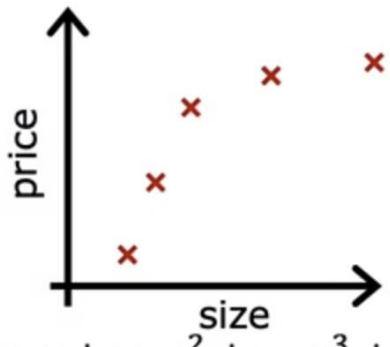




$w_1 x + b$

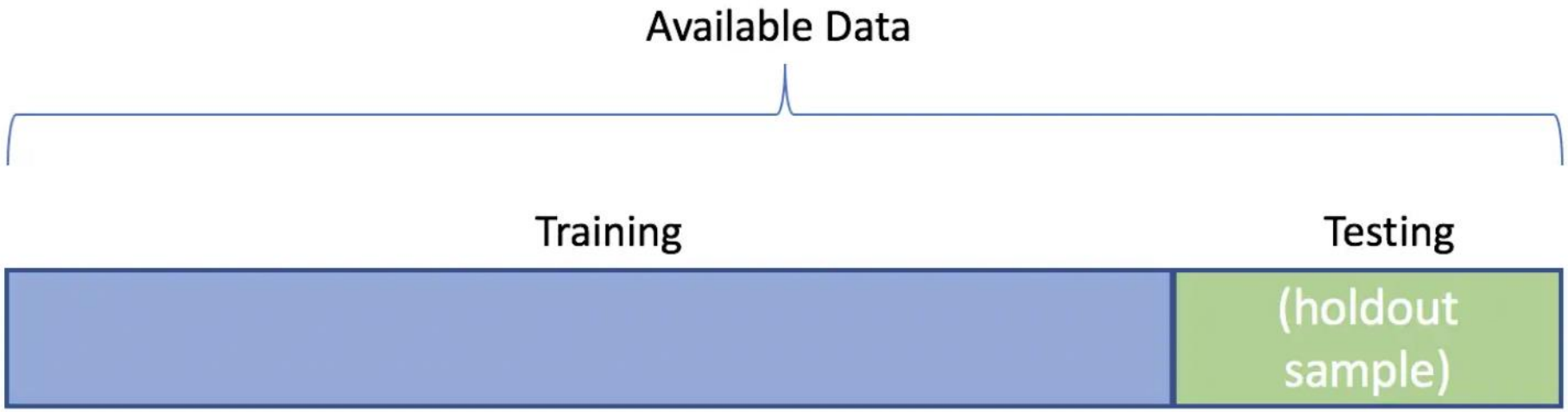


$w_1 x + w_2 x^2 + b$



$w_1 x + w_2 x^2 + w_3 x^3 + w_4 x^4 + b$

Measuring Overfitting



Exercise 4: Measuring Overfitting

8.2 Measuring Overfitting

We now split the data set in training data set and validation data set.

```
#Permutate the data
np.random.seed(123)
perm = np.random.permutation(X_train_extended.shape[0])
X_train_extended = X_train_extended[perm,:]
y_train = y_train[perm]

#Split the data
size_train = 66
X_split_train, y_split_train = X_train_extended[0:size_train,:], y_train[0:size_train]
X_split_val, y_split_val = X_train_extended[size_train:,:], y_train[size_train:]
```

Exercise 4: Measuing Training and Test Error

- Add polynomial features using the code above. Start with degree=2 and move to degree=20.
- Look how the Cost function for the training and the test cost evolves.
- Increase and decrease the number of iterations gradient descent is run (move it from 10000 to 20000) and observed what happen with the the training and the test cost.
- What do you think is happening?
- Which should be the degree of the included polynomial features?

Exercise 5: Vectorization

Exercise 5. Vectorize Linear Regression with Multiple Variables

In this exercise, you will have to vectorized the methods that appears with the heading **Vectorized**. Vectorization using NumPy allows for significantly faster computations compared to explicit loops, especially when dealing with large datasets. This is because NumPy operations are optimized and can leverage underlying C implementations for efficiency.

Specifically you have to vectorize the following methods:

- **Section 4** --> def compute_cost(X, y, w, b):
- **Section 5** --> def compute_gradient(X, y, w, b):

4 [VECTORIZE] Compute Cost With Multiple Variables

The equation for the cost function with multiple variables $J(\mathbf{w},b)$ is:

$$J(\mathbf{w},b) = \frac{1}{2m} \sum_{i=0}^{m-1} (f_{\mathbf{w},b}(\mathbf{x}^{(i)}) - y^{(i)})^2 \tag{3}$$

where:

$$f_{\mathbf{w},b}(\mathbf{x}^{(i)}) = \mathbf{w} \cdot \mathbf{x}^{(i)} + b \tag{4}$$

In contrast to tje previous exercise, $\mathbf{w}$ and $\mathbf{x}^{(i)}$ are vectors rather than scalars supporting multiple features.

```python
def compute_cost(X, y, w, b):
    """
    compute cost
    Args:
      X (ndarray (m,n)): Data, m examples with n features
      y (ndarray (m,)) : target values
      w (ndarray (n,)) : model parameters
      b (scalar)       : model parameter

    Returns:
      cost (scalar): cost
    """
    #### VECTORIZE ####
    return ?????
    ##################
```

Kaggle Competition

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Overview

Description

Evaluation

Bike sharing systems are a means of renting bicycles where the process of obtaining membership, rental, and bike return is automated via a network of kiosk locations throughout a city. Using these systems, people are able rent a bike from a one location and return it to a different place on an as-needed basis. Currently, there are over 500 bike-sharing programs around the world.

The data generated by these systems makes them attractive for researchers because the duration of travel, departure location, arrival location, and time elapsed is explicitly recorded. Bike sharing systems therefore function as a sensor network, which can be used for studying mobility in a city. In this competition, participants are asked to combine historical usage patterns with weather data in order to forecast bike rental demand in the Capital Bikeshare program in Washington, D.C.